

Academic Year 2024-25 (Odd Semester)

Course	:	B.Tech.	Date	:	9/11/2024
Semester/ Section	:	I/ L,M,N,O,P,Q,R,S,T,U,V	Time Duration	:	1:30 Hrs.
Subject & Subject Code	:	Programming for Problem Solving (1FY3-06)	Max. Marks	:	50

Course Outcomes

CO1	Understand concept of low-level and high-level languages, primary and secondary memory. Represent algorithm through flowchart and pseudo code for problem solving.
CO2	Represent and convert numbers & alphabets in various notations.

Q. No.	CO	Questions	Marks
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PART- A:

Attempt All Questions (10x2 = 20 Marks)

1.	1	Define PSEUDO code with an example.	2
2.	1	What is the difference between flowchart and algorithm?	2
3.	1	Explain Primary memory with example.	2
4.	1	Explain the Memory Access Methods.	2
5.	1	What is compiler?	2
6.	2	Find the Hexadecimal equivalent of the number $(1101011.110)_2$	2
7.	2	Find the Octal equivalent of the number $(B2A1.6C)_{16}$.	2
8.	2	Convert $(39.B8)_{16}$ to its decimal equivalent.	2
9.	2	Write the ASCII-7 binary notation for the word "Jaipur".	2
10.	2	Find the 1's and 2's complement of $(10110111)_2$ and add them.	2

PART-B: Descriptive /Analytical / Problem Solving/ Design questions
Attempt Any Three Questions (3x5=15 Marks)

1.	1	Illustrate the difference between High-Level language & Low- Level language.	5
2.	1	Categorize & Explain the types of memory.	5
3.	1	Explain the Von Neumann Architecture with a diagram.	5
4.	1	Write an algorithm and draw a flowchart to find the greatest of 2 numbers. }	5

PART-C: Descriptive /Analytical / Problem Solving/ Design questions
Attempt Any Three Questions (3x5=15 Marks)

1.	2	Solve the following: - (i) Multiply $(1111011)_2$ with $(1110)_2$ (ii) Divide $(10111011)_2$ by $(110)_2$	5
2.	2	Write the ASCII-7 & ASCII-8 Code for your College Name in both binary & hexadecimal notations.	5
3.	2	Convert the following: (i) $(1998)_{10} = (?)_2$ (ii) $(11011)_2 = (?)_{10}$ (iii) $(921)_{10} = (?)_8$	5
4.	2	Explain the followings with suitable example: (a) Binary addition (b) Binary subtraction	5